Security Operations Center (SOC) Internship Report

Introduction

A Security Operations Center (SOC) is a centralized unit responsible for monitoring, detecting, and responding to cybersecurity threats in an organization. SOC teams operate continuously to ensure system security and protect sensitive data from attacks.
SOC analysts use tools such as:

- Splunk
- Elastic SIEM
- Wireshark

They follow frameworks like:

- National Institute of Standards and Technology (NIST)
- MITRE ATT&CK

SOC Fundamentals and Operations

Purpose of SOC

- Detect cyber threats proactively
- Monitor systems continuously
- Respond to incidents quickly

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SOC Manager
      |
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 |      |          |
Tier 1  Tier 2   Tier 3
(Alert  (Invest. (Threat
Monitor) Analyst) Hunting)
```

SOC Roles

- Tier 1 Analyst: Monitors alerts
- Tier 2 Analyst: Investigates incidents
- Tier 3 Analyst: Performs threat hunting
- SOC Manager: Oversees operations

Key Functions

- Log analysis
- Alert triage
- Threat intelligence integration

SOC Workflow Diagram

Logs / Alerts
↓
Detection
↓
Triage
↓
Investigation
↓
Response

Security Monitoring Basics

Security monitoring involves tracking system and network activities to detect:

- Unauthorized access
- Malware activity
- Policy violations

Tools Used

- SIEM tools (Splunk, Elastic)
- Network analyzers like Wireshark

Key Metrics

- False Positives
- False Negatives
- Mean Time to Detect (MTTD)

Systems / Network
↓
Log Collection
↓
SIEM
↓
Alerts Generated
↓
SOC Analyst

Log Management Fundamentals

Logs are records of system activities.

- Log Lifecycle
- Collection
- Normalization
- Storage
- Analysis
- Log Types
- Windows Event Logs
- Syslog
- HTTP logs

Basic Security Concepts

CIA Triad Diagram

```
                Confidentiality
                   /       \
                  /         \
                 /           \
Integrity -------- Availability
```

Incident Response Lifecycle

Based on National Institute of Standards and Technology:

- Preparation
- Identification
- Containment
- Eradication
- Recovery
- Lessons Learned

Log Analysis (Windows Event Viewer)

Steps:
Open Event Viewer
Go to Windows Logs → Security
Filter:
Event ID = 4625
Identify failed login attempts
Result:
Multiple failed logins indicate a brute-force attack.

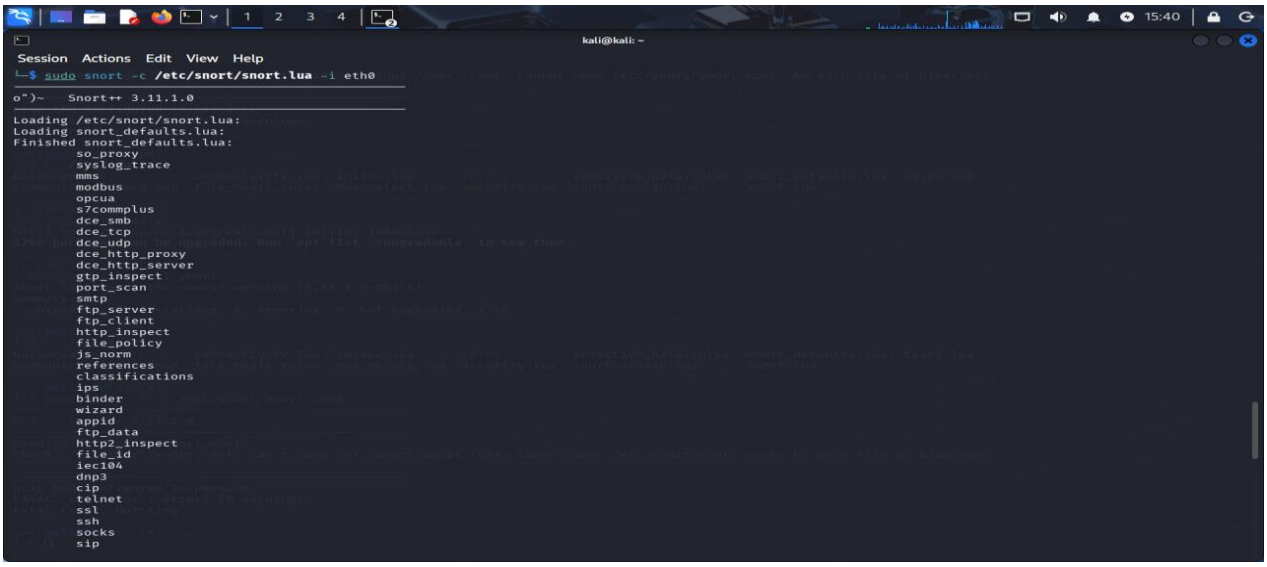
Snort Rule Testing

Tool: Snort

Steps:

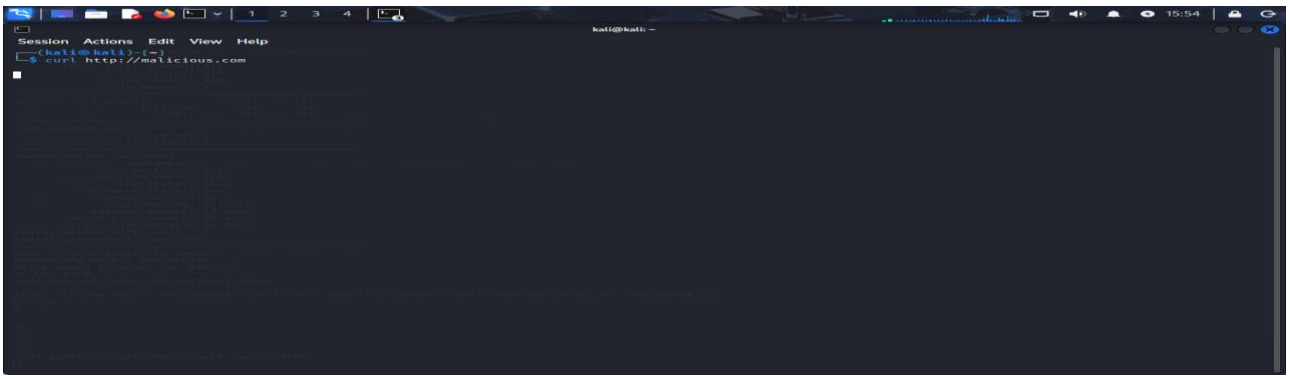
Install Snort
Add rule in local.rules
Run Snort

Rule Used: alert tcp any any -> any 80 (msg:"Malicious Domain"; content:"malicious.com"; http uri; sid:1000001;)



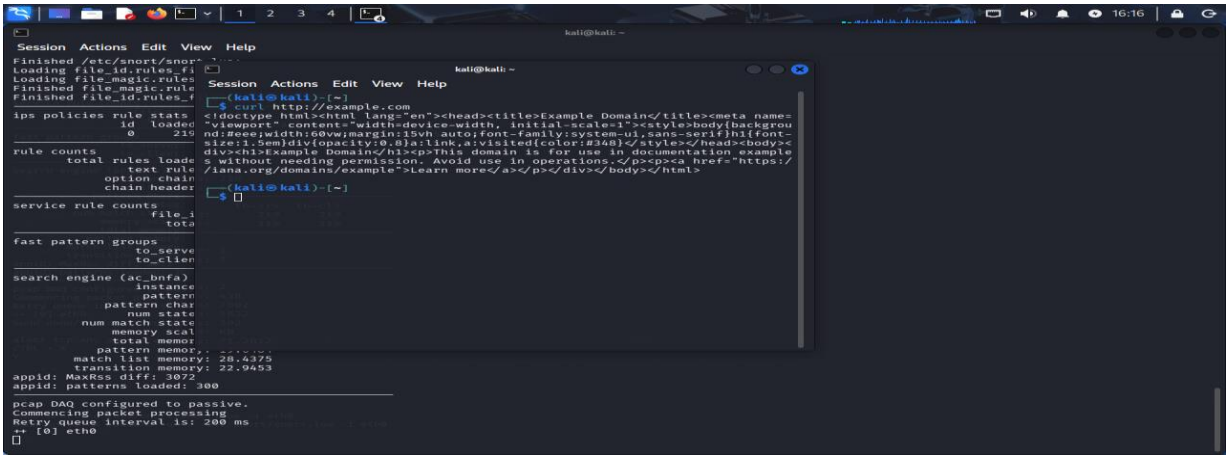
Test using:

curl http://malicious.com



Result:

Snort successfully detected malicious traffic.



Nessus Vulnerability Scan

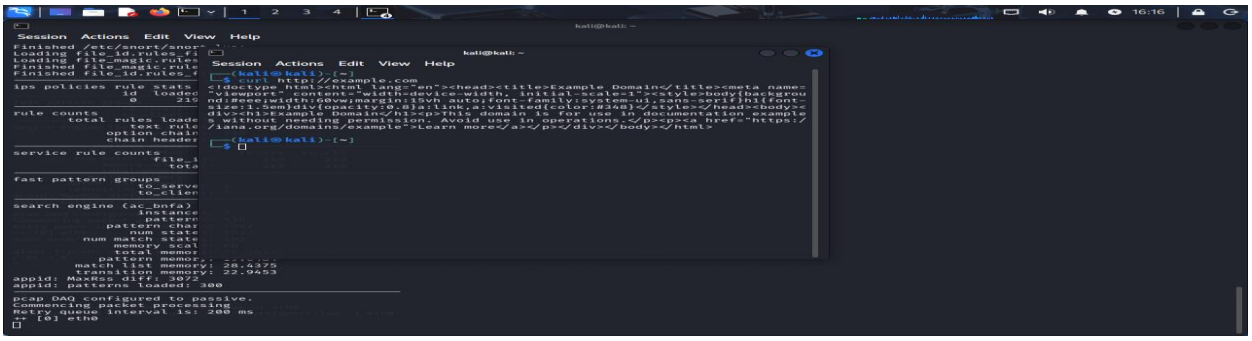
Tool: Nessus

Steps:

- Install Nessus Essentials
- Scan Metasploitable VM
- Export report

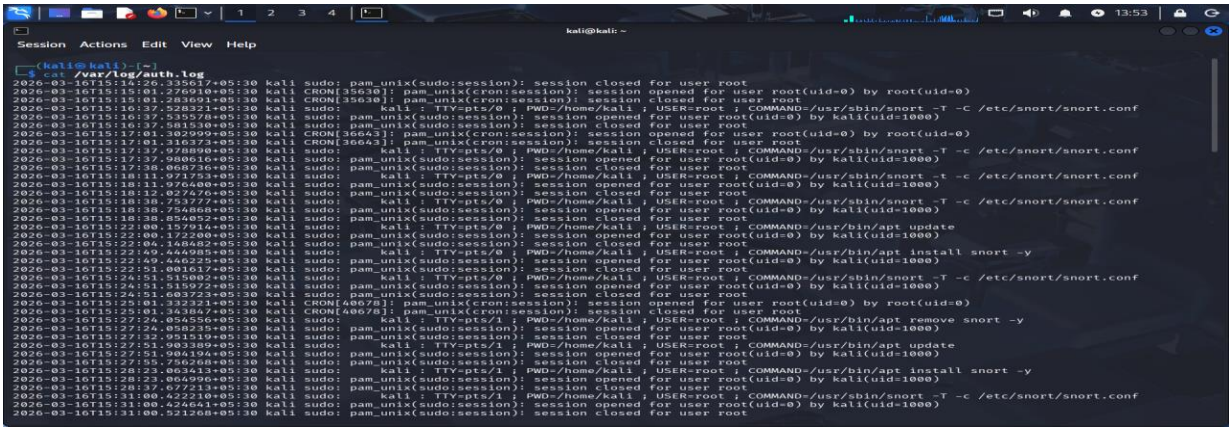
Result:

Top vulnerabilities identified based on CVSS score.



Log Collection Pipeline

Steps:
Install Fluentd on Ubuntu
Generate log:
logger "Test message"
Forward logs to Elastic SIEM
Result:
Logs successfully collected and analyzed.

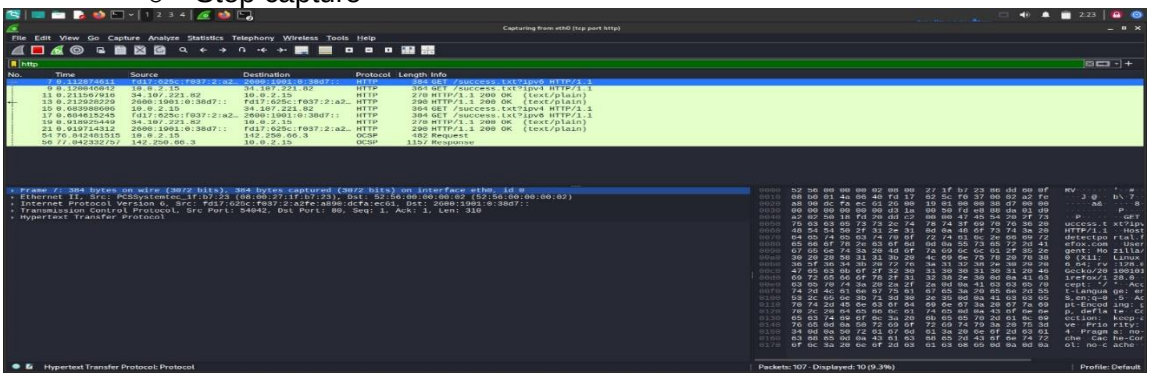


Network Monitoring Using Wireshark

Open: Wireshark

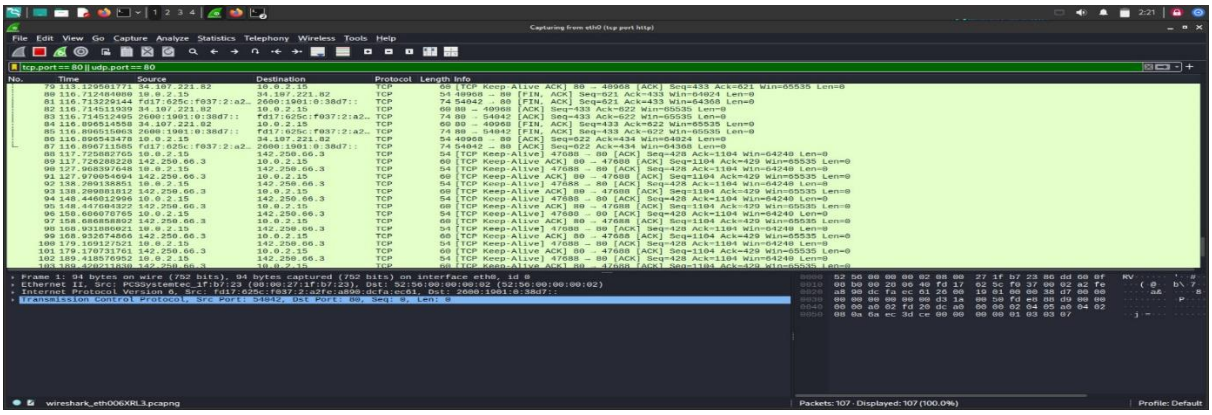
Steps:

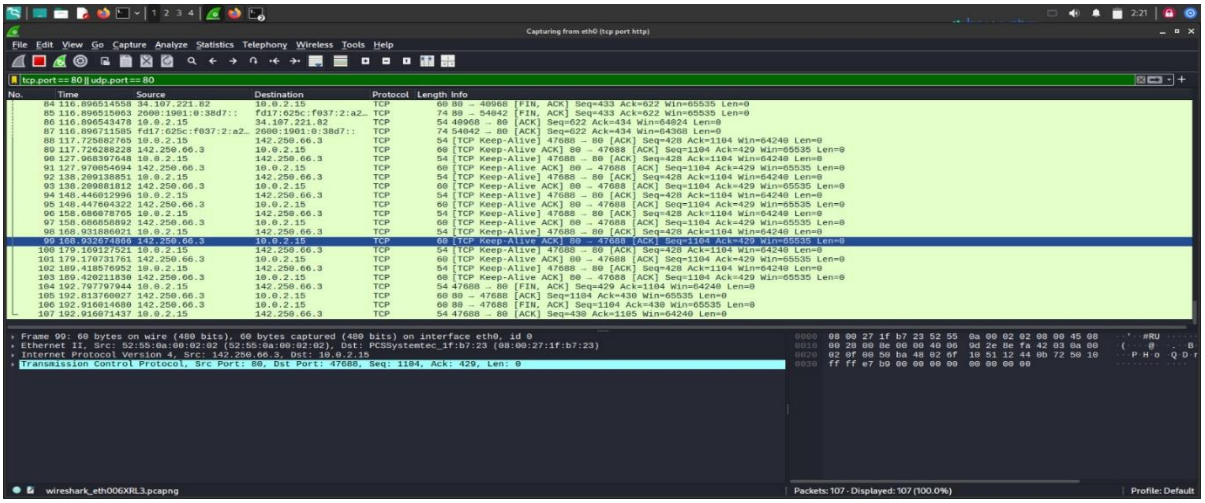
- Select eth0 interface
- Click Start Capture
- Generate traffic by opening websites
- Stop capture



Filter example:

http





Dashboard Creation

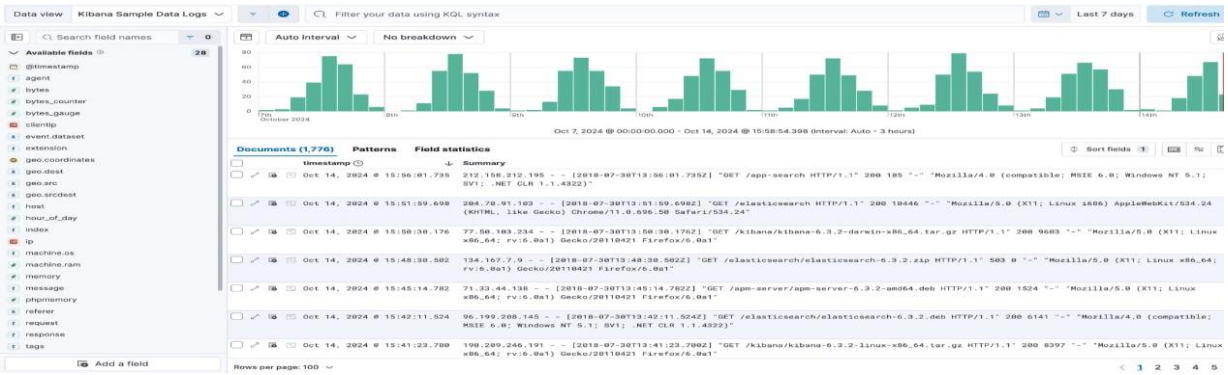
Tools: Kibana

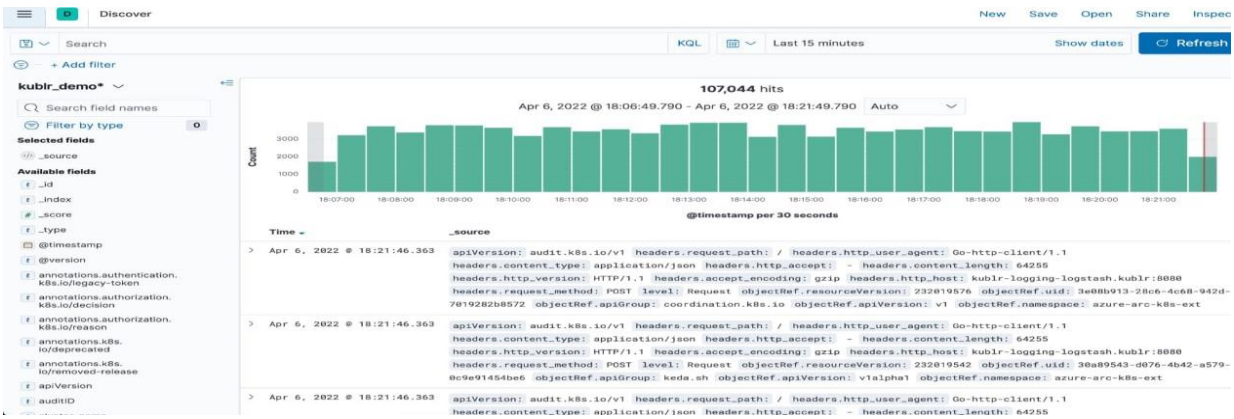
Dashboard Metrics:

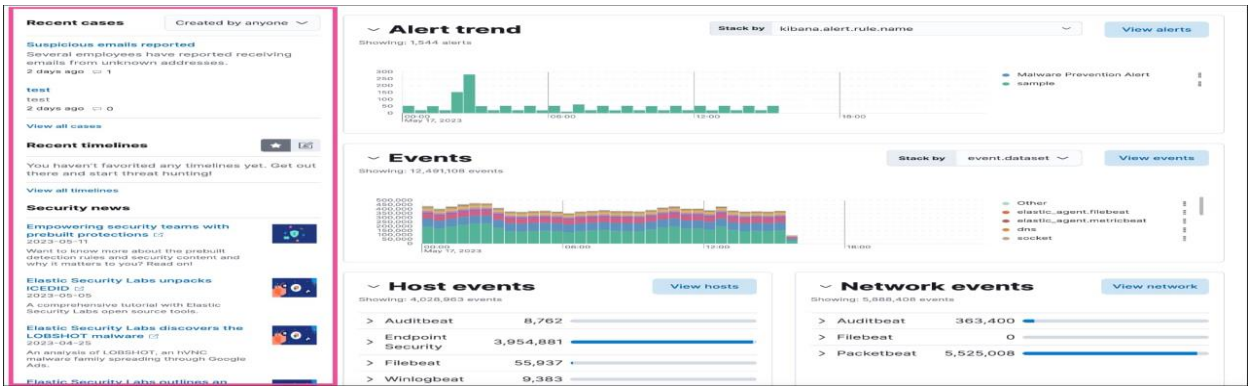
Top attacker IPs

Event frequency

Alert severity







The knowledge gained includes:

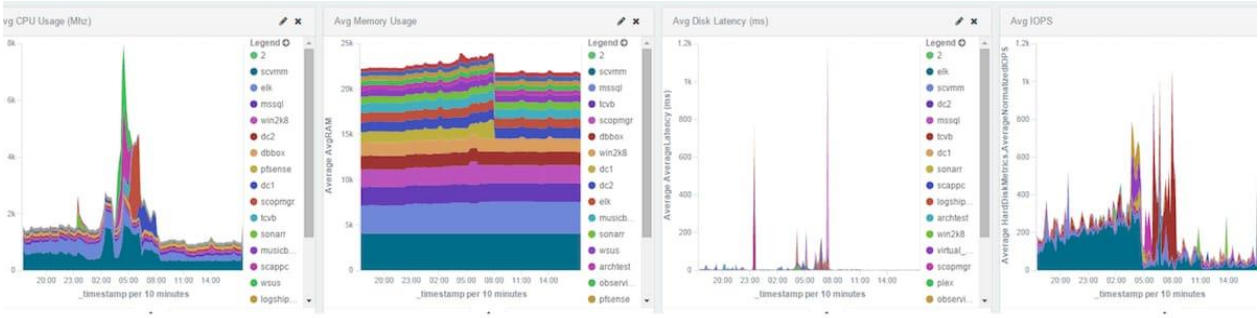
Log analysis

Threat detection

Incident response

Alert configuration

This training is essential for building a career as a SOC Analyst.



Conclusion:

This SOC training task provided practical exposure to real-world cybersecurity operations. The implementation of tools like Snort, Wireshark, and SIEM platforms helped in understanding how security analysts detect threats and respond to incidents.